		≈ 6.3 J
10	**B**	Force on satellite $F = \frac{d(mv)}{dt}$ $= v\frac{dm}{dt}$ $= 2000 \times 0.20$ $= 400$ N Final mass of satellite after 5.0 s $m_f = 400 - (0.20)(5.0) = 399$ kg

No.	Answer	Solution
		Impulse experienced by satellite $F \times t = m_f v - m_i u$ $(400)(5.0) = 399v - (400)(3000)$ $v = 3013$ $\approx 3010 \text{ m s}^{-1}$
11	B	Extension of spring with load of 6.0 N is 0.020 m. By extending a further